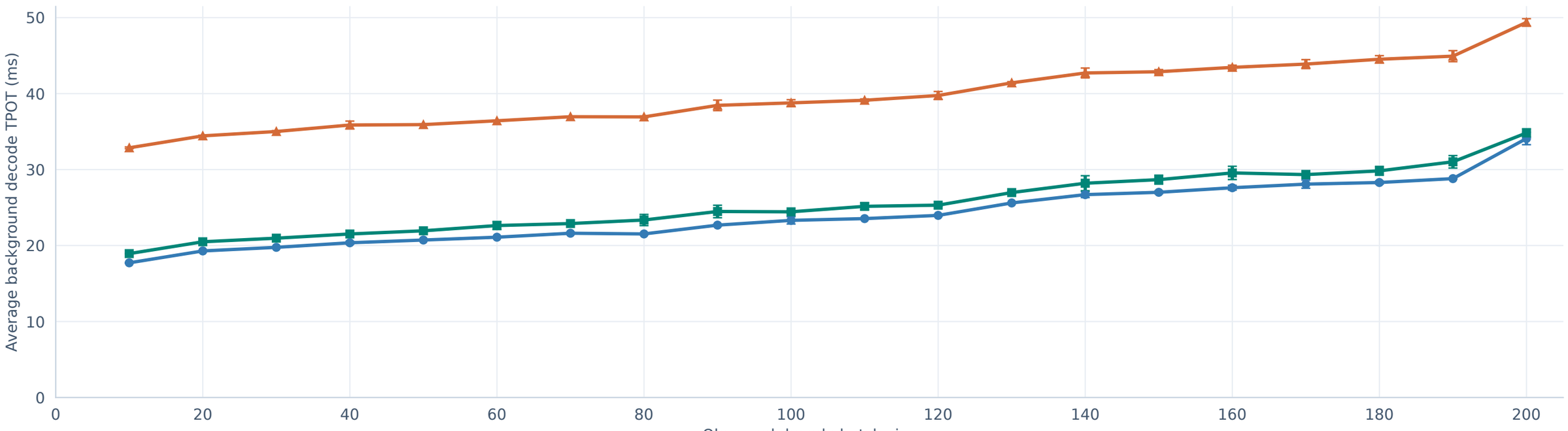
PHASE 8 / SAME-CONTEXT FULL VS APPEND / 16K HISTORY

Background decode TPOT under prefill interference

Single RTX 4090 | Qwen2.5-7B | 128 output tokens per background request | Mean and sample SD, 5 repeats



For each background request: TPOT = (last token arrival - first token arrival) / 127. Average requests within each trial, then five repeats.

Only background decode requests contribute to these curves. Interferer TTFT and prefill model wall time are separate measurements.

Adapted workload: independent 128-token histories; full recomputes 17,408 tokens, append computes 1,024 with 16,384 cached. Batch 0 has no TPOT.